

7. Integration Techniques

7.4. Integration by Partial Fractions

7.4a Learning Objectives

- I can understand how to rewrite rational functions as a sum of partial fractions.
- I can understand when and how to integrate by partial fractions.

7.4.1. Motivation

How can we rewrite the function $f(x) = \frac{1}{x-4} - \frac{1}{x-5}$ ($f(x)$ equals 1 over $x - 4$ minus 1 over $x - 5$)?

$$\begin{aligned} f(x) &= \frac{1}{x-4} - \frac{1}{x-5} \\ &= \frac{x-5}{(x-4)(x-5)} - \frac{x-4}{(x-4)(x-5)} \\ &= \frac{x-5-x+4}{(x-4)(x-5)} \\ &= \frac{-1}{(x-4)(x-5)} \end{aligned}$$

$f(x)$ equals 1 over $x - 4$ minus 1 over $x - 5$ equals $x - 5$ over $(x - 4)(x - 5)$ minus $x - 4$ over $(x - 4)(x - 5)$ equals $x - 5 - x + 4$ over $(x - 4)(x - 5)$ equals -1 over $(x - 4)(x - 5)$

Which is easier to take the integral of:

$$\frac{1}{x-4} - \frac{1}{x-5} \quad \text{or} \quad \frac{-1}{(x-4)(x-5)}?$$

1 over $x - 4$ minus 1 over $x - 5$ or -1 over $(x - 4)(x - 5)$?

The difference on the left is easier to take the integral of.

So, if we are asked to integrate the expression on the right, it would be nice to rewrite it as the expression on the left first. This process of rewriting a rational function as a sum of “simpler” rational functions is called partial fraction decomposition.

7.4.2. Partial Fraction Decomposition

Theorem 7.4.1 Fundamental Theorem of Algebra

If $P(x)$ (P of x) is a polynomial with real coefficients, then $P(x)$ can be factored into linear $(ax + b)$ (a x plus b) and irreducible quadratic $(ax^2 + bx + c)$ (a x squared plus b x plus c) factors.

$$P(x) = \underbrace{(a_1x + b_1) \dots (a_nx + b_n)}_{\text{linear}} \underbrace{(A_1x^2 + B_1x + C_1) \dots (A_mx^2 + B_mx + C_m)}_{\text{quadratic}}$$

P of x equals a one x plus b one times dot dot dot times a n x plus b n times a one x squared plus b one x plus c one times dot dot dot a m x squared plus b m x plus c m .

Idea:

Say $f(x) = \frac{P(x)}{Q(x)}$ ($f(x)$ equals $P(x)$ over $Q(x)$) where $P(x)$ and $Q(x)$ are polynomials. Then, we can factor the denominator, $Q(x)$ ($Q(x)$), into its linear and quadratic terms:

$$Q(x) = (a_1x + b_1) \dots (a_nx + b_n)(A_1x^2 + B_1x + C_1) \dots (A_mx^2 + B_mx + C_m)$$

$Q(x)$ equals $(a_1x + b_1) \dots (a_nx + b_n)(A_1x^2 + B_1x + C_1) \dots (A_mx^2 + B_mx + C_m)$

Then we can write $f(x)$ ($f(x)$) as:

$$f(x) = S(x) + \frac{J_1}{a_1x + b_1} + \dots + \frac{J_2}{a_nx + b_n} + \frac{K_1}{A_1x^2 + B_1x + C_1} + \dots + \frac{J_2}{A_mx^2 + B_mx + C_m}$$

$f(x)$ equals $S(x)$ plus J_1 over $a_1x + b_1$ plus $\dots$ plus J_2 over $a_nx + b_n$ plus K_1 over $A_1x^2 + B_1x + C_1$ plus $\dots$ plus J_2 over $A_mx^2 + B_mx + C_m$

where $S(x)$ ($S(x)$) is a polynomial.

More precisely, the steps you can use are as follows.

Tip

General process for decomposing a rational function into its partial fractions.

1. If $f(x) = \frac{P(x)}{Q(x)}$ ($f(x)$ equals $P(x)$ over $Q(x)$) is not *proper* ($\deg P(x) \geq \deg Q(x)$ ($\deg$ of $P(x)$ is greater than or equal to $\deg$ of $Q(x)$)) then use polynomial long division to find

$$f(x) = S(x) + \frac{R(x)}{Q(x)}, \text{ with } \deg R(x) < \deg Q(x)$$

$f(x)$ equals $S(x)$ plus $R(x)$ over $Q(x)$, with $\deg$ of $R(x)$ is less than $\deg$ of $Q(x)$

2. Then, if $\frac{R(x)}{Q(x)}$ ($R(x)$ over $Q(x)$) cannot be integrated as is, factor $Q(x)$ ($Q(x)$) into linear and quadratic irreducible terms.
3. Split $\frac{R(x)}{Q(x)}$ ($R(x)$ over $Q(x)$) into its sum of partial fractions with unknown numerators.

Factor in $Q(x)$ ($Q(x)$)	Sum of Partial Fractions
$(ax + b)^m$ $(ax + b)^m$	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \dots + \frac{A_m}{(ax + b)^m}$ A sub 1 over a x plus b plus A sub 2 over (a x plus b) squared plus ... plus A sub m over (a x plus b) m
$(ax^2 + bx + c)^n$ $(ax^2 + bx + c)^n$	$\frac{B_1x + C_1}{ax^2 + bx + c} + \frac{B_2x + C_2}{(ax^2 + bx + c)^2} + \dots$ $+ \frac{B_mx + C_m}{(ax^2 + bx + c)^m}$ B sub 1 x plus C sub 1 over a x squared plus b x plus c plus B sub 2 x plus C sub 2 over (a x squared plus b x plus c) squared plus ... plus B sub m x plus C sub m over (a x squared plus b x plus c) m

4. Solve for the numerators.

1. Set original fraction equal to the sum of partial fractions.
2. Multiply both sides by $Q(x)$ ($Q(x)$).
3. Solve for unknown constants using a system of equations and/or plugging in some x (x) values.

Example 7.4.1

Use polynomial long division to rewrite

$$f(x) = \frac{x^4 + 3x^3 + 5x + 6}{x + 2}$$

$f(x)$ equals x to the power of 4 plus 3 x cubed plus 5 x plus 6 over x plus 2

then evaluate

$$\int f(x) dx$$

integral of $f(x)$ dx

Solution

So,

$$\frac{x^4 + 3x^3 + 5x + 6}{x + 2} = (x^3 + x^2 - 2x + 9) + \frac{-12}{x + 2}$$

x to the power of 4 plus 3 x cubed plus 5 x plus 6 over x plus 2 equals (x cubed plus x squared $- 2x$ plus 9) plus $- 12$ over x plus 2

Then, with a quick and small u (u)-sub,

$$\begin{aligned}\int \frac{x^4 + 3x^3 + 5x + 6}{x + 2} dx &= \int (x^3 + x^2 - 2x + 9) dx + \int \frac{-12}{x + 2} dx \\ &= \frac{x^4}{4} + \frac{x^3}{3} - x^2 + 9x - 12 \ln|x + 2| + C\end{aligned}$$

integral of x to the power of 4 plus 3 x cubed plus 5 x plus 6 over x plus 2 d x equals
integral of (x cubed plus x squared $- 2x$ plus 9) d x plus integral of $- 12$ over x plus 2 d
 x equals x to the power of 4 over 4 plus x cubed over 3 $- x$ squared plus 9 $x - 12 \ln$ of | x
plus 2 | plus C

Example 7.4.2

Set up the partial fraction decomposition for this rational function, but you do not need to find the coefficients of the terms in the numerators.

$$\frac{P(x)}{(x - 1)^3(2x - 3)(x^2 + 2x + 15)^2}$$

$P(x)$ over ($x - 1$) cubed ($2x - 3$) (x squared plus 2 x plus 15) squared

Solution

We first check the quadratic term is actually irreducible: since $b^2 - 4ac = 4 - 4(1)(15) < 0$ (b squared $- 4ac$ equals $4 - 4(1)(15)$ is less than 0), this is irreducible and we do not need to further factor it.

Then, we have

$$\begin{aligned}\frac{P(x)}{(x - 1)^3(2x - 3)(x^2 + 2x + 15)^2} = \\ \frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{(x - 1)^3} + \frac{D}{2x - 3} + \frac{Ex + F}{x^2 + 2x + 15} + \frac{Gx + H}{(x^2 + 2x + 15)^2}\end{aligned}$$

$P(x)$ over ($x - 1$) cubed ($2x - 3$) (x squared plus 2 x plus 15) squared equals A
over $x - 1$ plus B over ($x - 1$) squared plus C over ($x - 1$) cubed plus D over $2x - 3$
plus E x plus F over x squared plus 2 x plus 15 plus G x plus H over (x squared plus 2 x
plus 15) squared

Example 7.4.3

Do the partial fraction decomposition on the following rational function.

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)}$$

x squared plus $2x - 1$ over $x(2x - 1)(x + 2)$

Solution

Each factor in the denominator is linear and has highest power 1. So, the partial fractions are in the form

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

x squared plus $2x - 1$ over $x(2x - 1)(x + 2)$ equals A over x plus B over $2x - 1$ plus C over $x + 2$

and we need to determine A, B, C (A, B, C). Then,

$$\begin{aligned} (x(2x - 1)(x + 2)) \frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} &= \left(\frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2} \right) (x(2x - 1)(x + 2)) \\ \Rightarrow (x^2 + 2x - 1) &= A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1) \end{aligned}$$

$(x(2x - 1)(x + 2))$ x squared plus $2x - 1$ over $x(2x - 1)(x + 2)$ equals $(A$ over x plus B over $2x - 1$ plus C over $x + 2$) $(x(2x - 1)(x + 2)) \Rightarrow (x$ squared plus $2x - 1)$ equals $A(2x - 1)(x + 2)$ plus $Bx(x + 2)$ plus $Cx(2x - 1)$

Now we can carefully plug in values to help us find A, B, C (A, B, C). First, if $x = \frac{1}{2}$ (x equals $\frac{1}{2}$) notice

$$\begin{aligned} \left(\left(\frac{1}{2} \right)^2 + 2 \left(\frac{1}{2} \right) - 1 \right) &= A \left(2 \left(\frac{1}{2} \right) - 1 \right) \left(\left(\frac{1}{2} \right) + 2 \right) + B \left(\frac{1}{2} \right) \left(\left(\frac{1}{2} \right) + 2 \right) + C \left(\frac{1}{2} \right) \left(2 \left(\frac{1}{2} \right) - 1 \right) \\ \frac{1}{4} &= B \left(\frac{1}{2} \right) \left(\frac{5}{2} \right) \\ B &= \frac{1}{4} (2) \left(\frac{2}{5} \right) \\ B &= \frac{1}{5} \end{aligned}$$

$((\frac{1}{2})^2 + 2(\frac{1}{2}) - 1)$ equals $A(2(\frac{1}{2}) - 1)((\frac{1}{2}) + 2)$ plus $B(\frac{1}{2})(\frac{1}{2} + 2)$ plus $C(\frac{1}{2})(2(\frac{1}{2}) - 1)$
 $\frac{1}{4}$ equals $B(\frac{1}{2})(\frac{5}{2})$ B equals $\frac{1}{4}(2)(\frac{2}{5})$ B equals $\frac{1}{5}$

If $x = 0$ (x equals 0) notice

$$((0)^2 + 2(0) - 1) = A(2(0) - 1)((0) + 2) + B(0)((0) + 2) + C(0)(2(0) - 1)$$

$$-1 = A(1)(2)$$

$$A = -\frac{1}{2}$$

((0) squared plus 2 (0) - 1) equals A (2 (0) - 1) ((0) plus 2) plus B (0) ((0) plus 2) plus C (0) (2 (0) - 1) - 1 equals A (1) (2) A equals - 1 over 2

and if $x = -2$ (x equals - 2) we have

$$((-2)^2 + 2(-2) - 1) = A(2(-2) - 1)((-2) + 2) + B(-2)((-2) + 2) + C(-2)(2(-2) - 1)$$

$$-1 = C(-2)(-5)$$

$$C = -\frac{1}{10}$$

((- 2) squared plus 2 (- 2) - 1) equals A (2 (- 2) - 1) ((- 2) plus 2) plus B (- 2) ((- 2) plus 2) plus C (- 2) (2 (- 2) - 1) - 1 equals C (- 2) (- 5) C equals - 1 over 10

Then the partial fraction decomposition is

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{-\frac{1}{2}}{x} + \frac{\frac{1}{5}}{2x - 1} + \frac{-\frac{1}{10}}{x + 2}.$$

x squared plus 2 x - 1 over x (2 x - 1) (x plus 2) equals - 1 over 2 over x plus 1 over 5 over 2 x - 1 plus - 1 over 10 over x plus 2 .

7.4.3. Integrals Needing Partial Fraction Decomposition

Example 7.4.4

Find the antiderivative

$$\int \frac{2x^3 - x^2 + 4x}{x^4 + 4x^2} dx$$

integral of 2 x cubed - x squared plus 4 x over x to the power of 4 plus 4 x squared d x

Solution

First we factor and simplify everything we can

$$\begin{aligned} \int \frac{2x^3 - x^2 + 4x}{x^4 + 4x^2} dx &= \int \frac{x(2x^2 - x + 4)}{x^2(x^2 + 4)} dx \\ &= \int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx \end{aligned}$$

integral of 2 x cubed - x squared plus 4 x over x to the power of 4 plus 4 x squared d x
equals integral of x (2 x squared - x plus 4) over x squared (x squared plus 4) d x
equals integral of 2 x squared - x plus 4 over x (x squared plus 4) d x

Then we find the partial fraction decomposition of the integrand. Since the $x^2 + 4$ (x squared plus 4) factor has discriminant $0^2 - 4(1)(4) = -16 < 0$ (0 squared minus 4 times 1 times 4 equals minus 16 is less than 0) it is irreducible. Then, the form of the partial fraction decomposition is

$$\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

2 x squared minus x plus 4 over x (x squared plus 4) equals A over x plus B x plus C over x squared plus 4

Multiplying by the denominator gives us

$$(2x^2 - x + 4) = (x(x^2 + 4))\frac{A}{x} + (x(x^2 + 4))\frac{Bx + C}{x^2 + 4}$$

$$(2x^2 - x + 4) = (x^2 + 4)A + x(Bx + C)$$

(2 x squared minus x plus 4) equals (x (x squared plus 4)) A over x plus (x (x squared plus 4)) B x plus C over x squared plus 4 (2 x squared minus x plus 4) equals (x squared plus 4) A plus x (B x plus C)

We need a slightly new strategy for B, C (B , C) since the roots of $x^2 + 4$ (x squared plus 4) are complex, we will use this strategy for finding A (A) also, but it isn't necessary. Multiplying things out

$$(2x^2 - x + 4) = (x^2 + 4)A + x(Bx + C)$$

$$2x^2 - x + 4 = x^2 + 4 + Bx^2 + Cx$$

$$2x^2 - x + 4 = (1 + B)x^2 + Cx + 4$$

(2 x squared minus x plus 4) equals (x squared plus 4) A plus x (B x plus C) 2 x squared minus x plus 4 equals x squared plus 4 plus B x squared plus C x 2 x squared minus x plus 4 equals (1 plus B) x squared plus C x plus 4

Now lining up like powers on both sides we see that

$$\underline{2x^2} - x + 4 = \underline{(1 + B)x^2} + Cx + 4$$

$$\Rightarrow 2 = 1 + B$$

$$-x + 4 \text{ equals } Cx + 4 \Rightarrow 2 \text{ equals } 1 + B$$

So $B = 1$ (B equals 1), and

$$2x^2 - \underline{x} + 4 = (1 + B)x^2 + \underline{Cx} + 4$$

$$\Rightarrow -1 = C$$

$$2x^2 \text{ plus } 4 \text{ equals } (1 + B)x^2 \text{ plus } -1x \text{ plus } 4 \Rightarrow -1 \text{ equals } C$$

7.4a Section Summary: